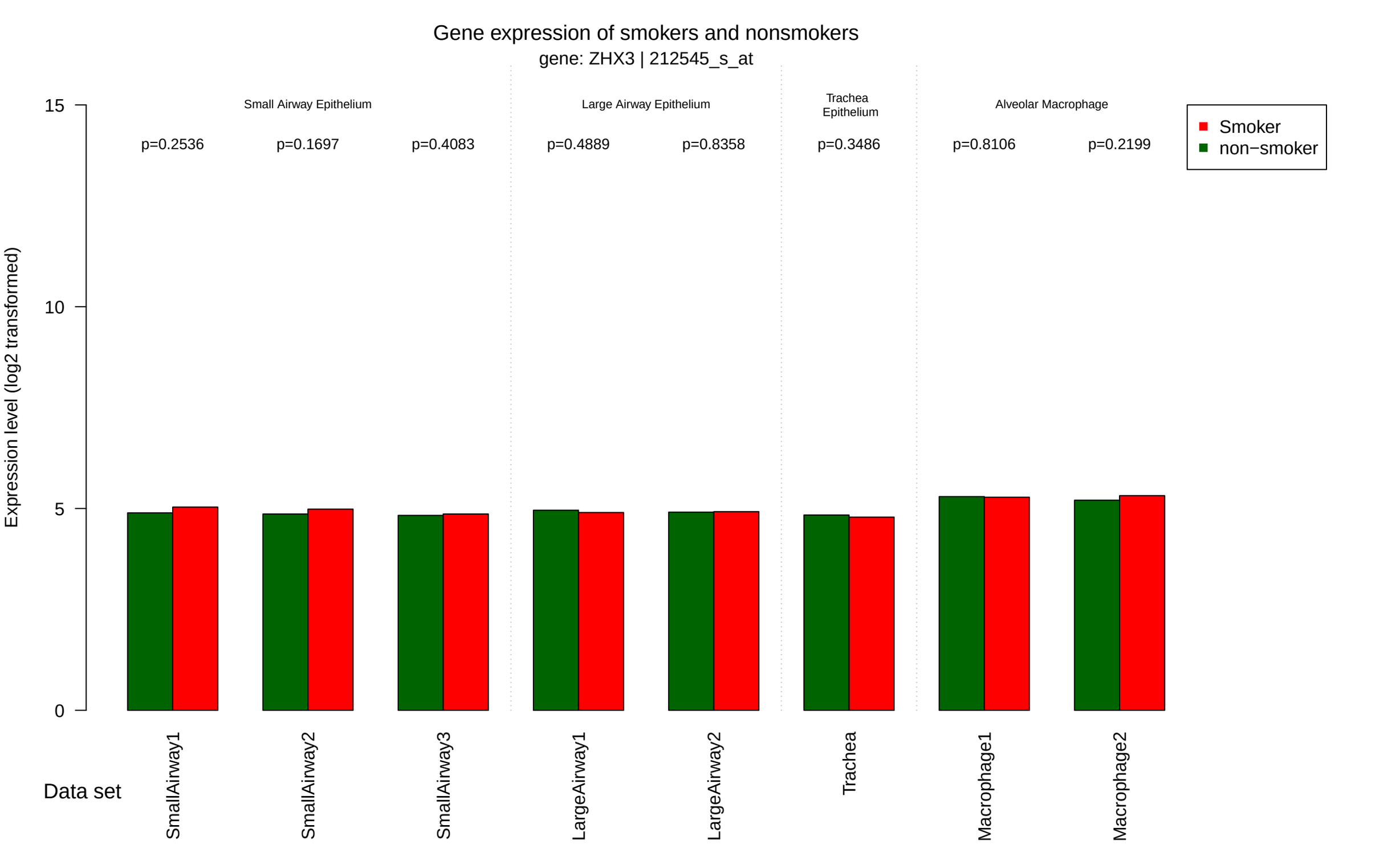
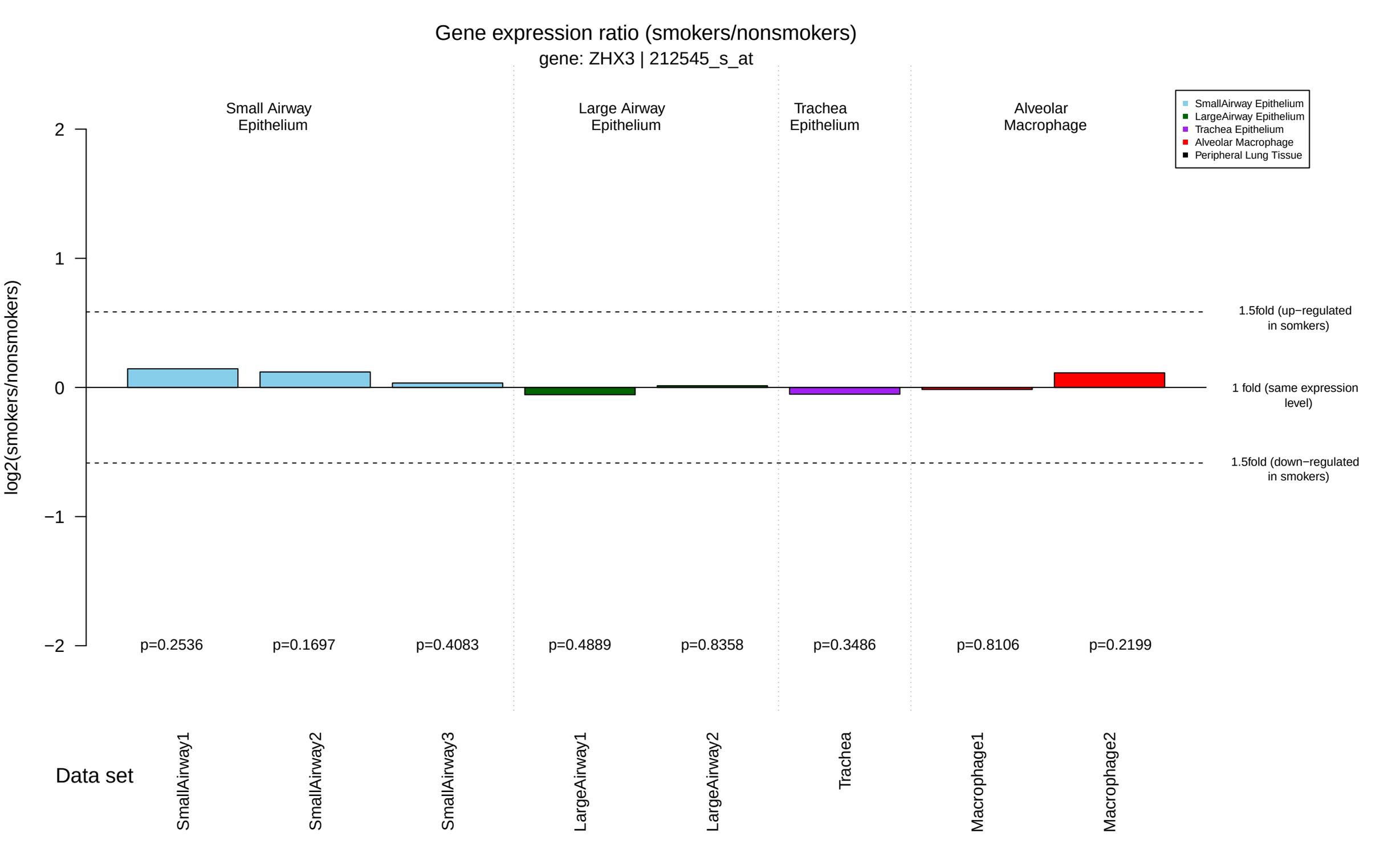
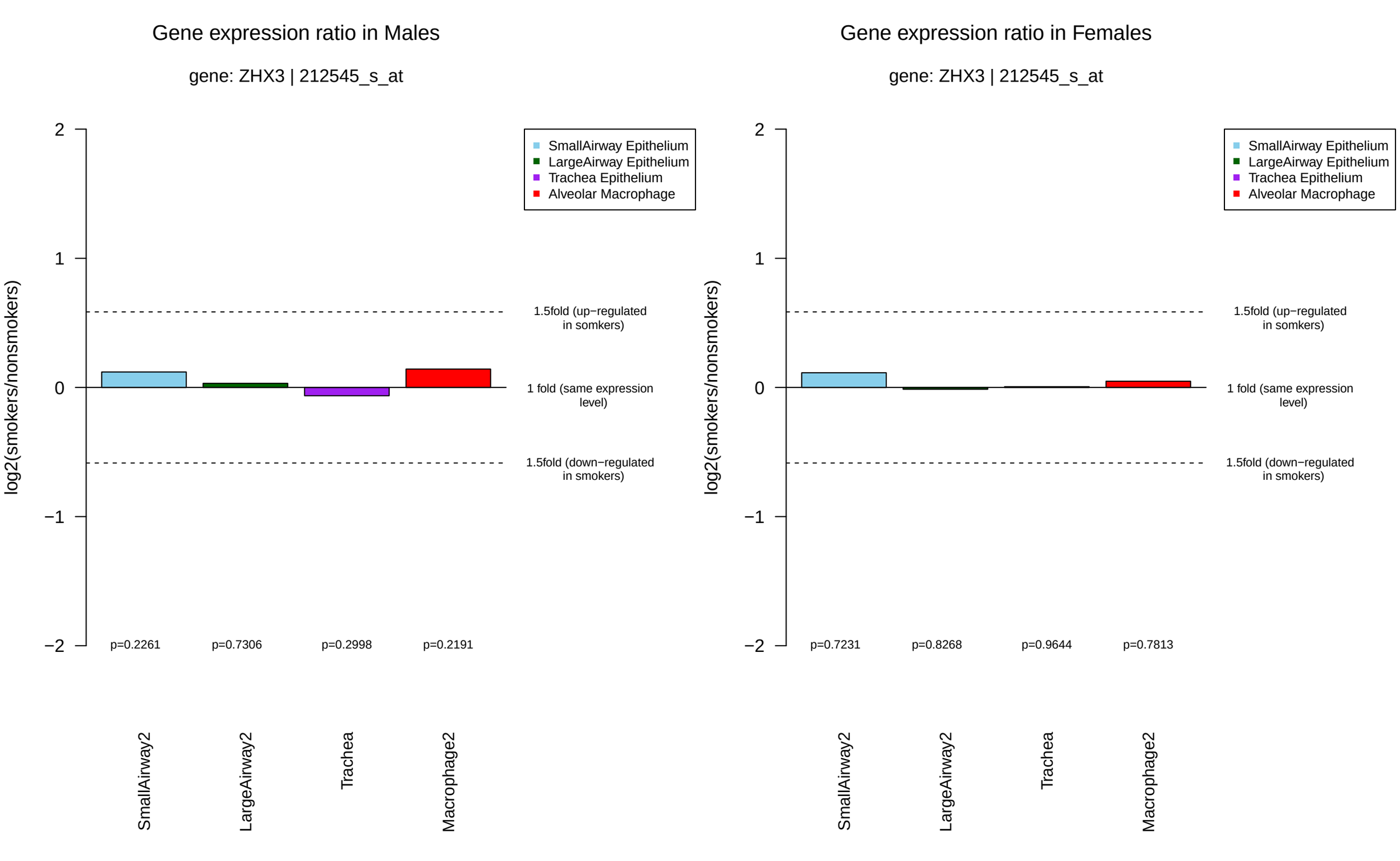
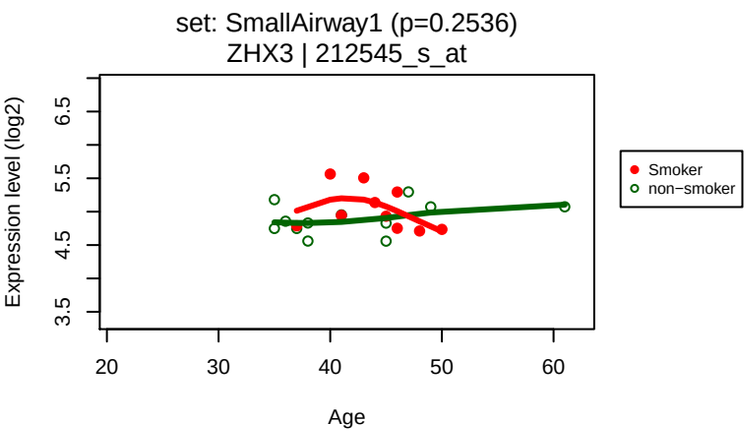
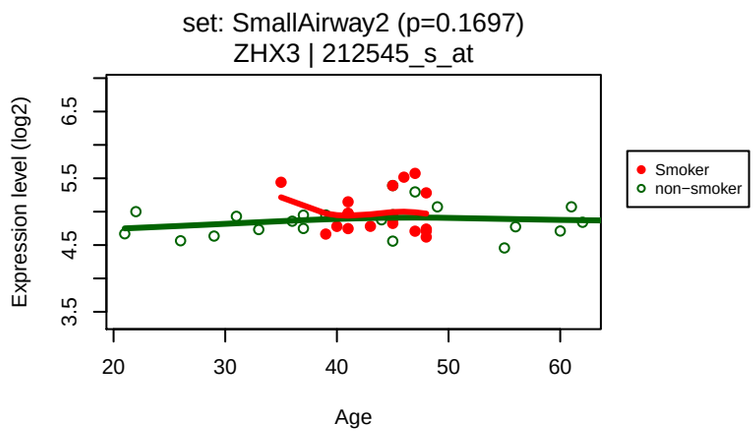
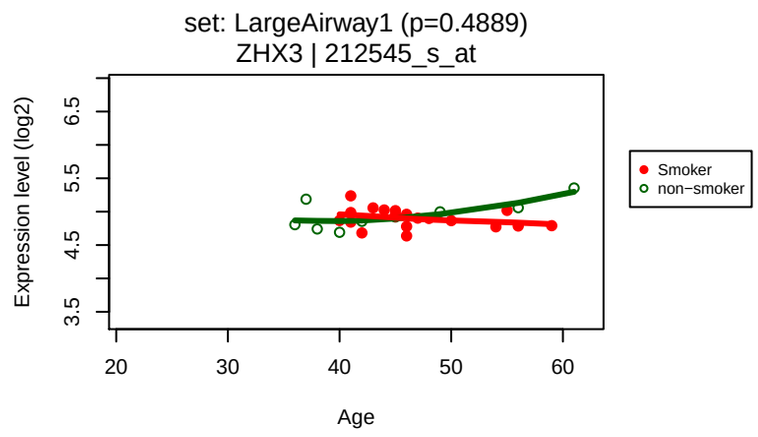

Gene expression of smokers and nonsmokers
gene: ZHX3 | 212545_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2536
p=0.1697
p=0.4083
p=0.4889
p=0.8358
p=0.3486
p=0.8106
p=0.2199
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

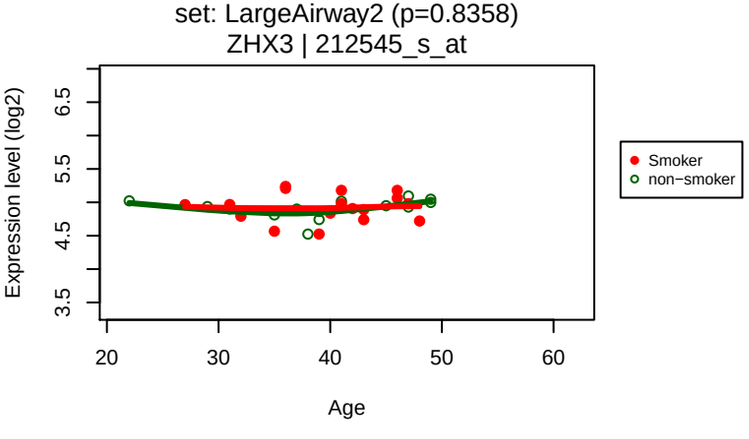

Gene expression ratio (smokers/nonsmokers)
gene: ZHX3 | 212545_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2536
p=0.1697
p=0.4083
p=0.4889
p=0.8358
p=0.3486
p=0.8106
p=0.2199
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

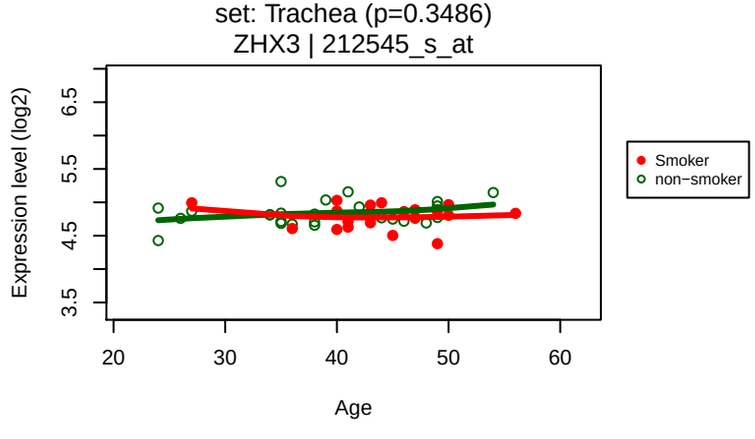

Gene expression ratio in Males
gene: ZHX3 | 212545_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2261
p=0.7306
p=0.2998
p=0.2191
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ZHX3 | 212545_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7231
p=0.8268
p=0.9644
p=0.7813
SmallAirway2
LargeAirway2
Trachea
Macrophage2

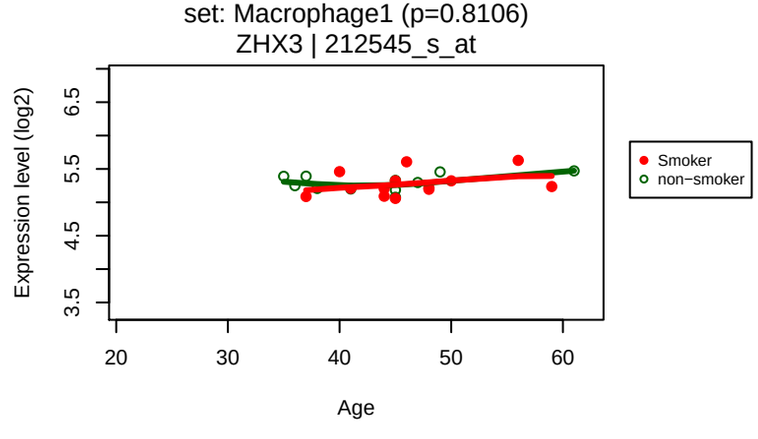

set: SmallAirway1 (p=0.2536)
ZHX3 | 212545_s_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker

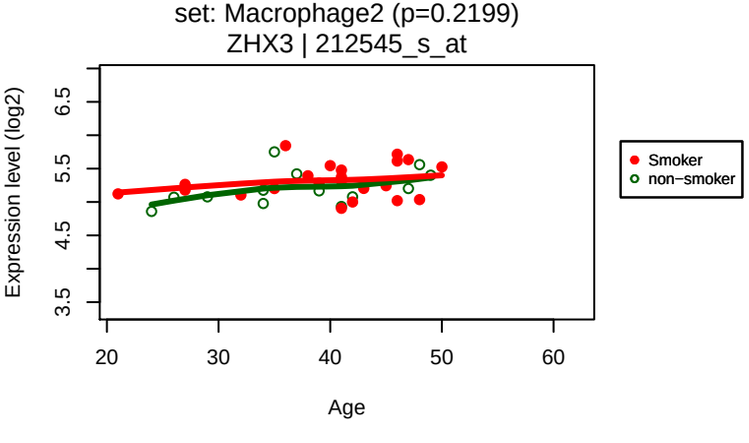

set: SmallAirway2 (p=0.1697)
ZHX3 | 212545_s_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker

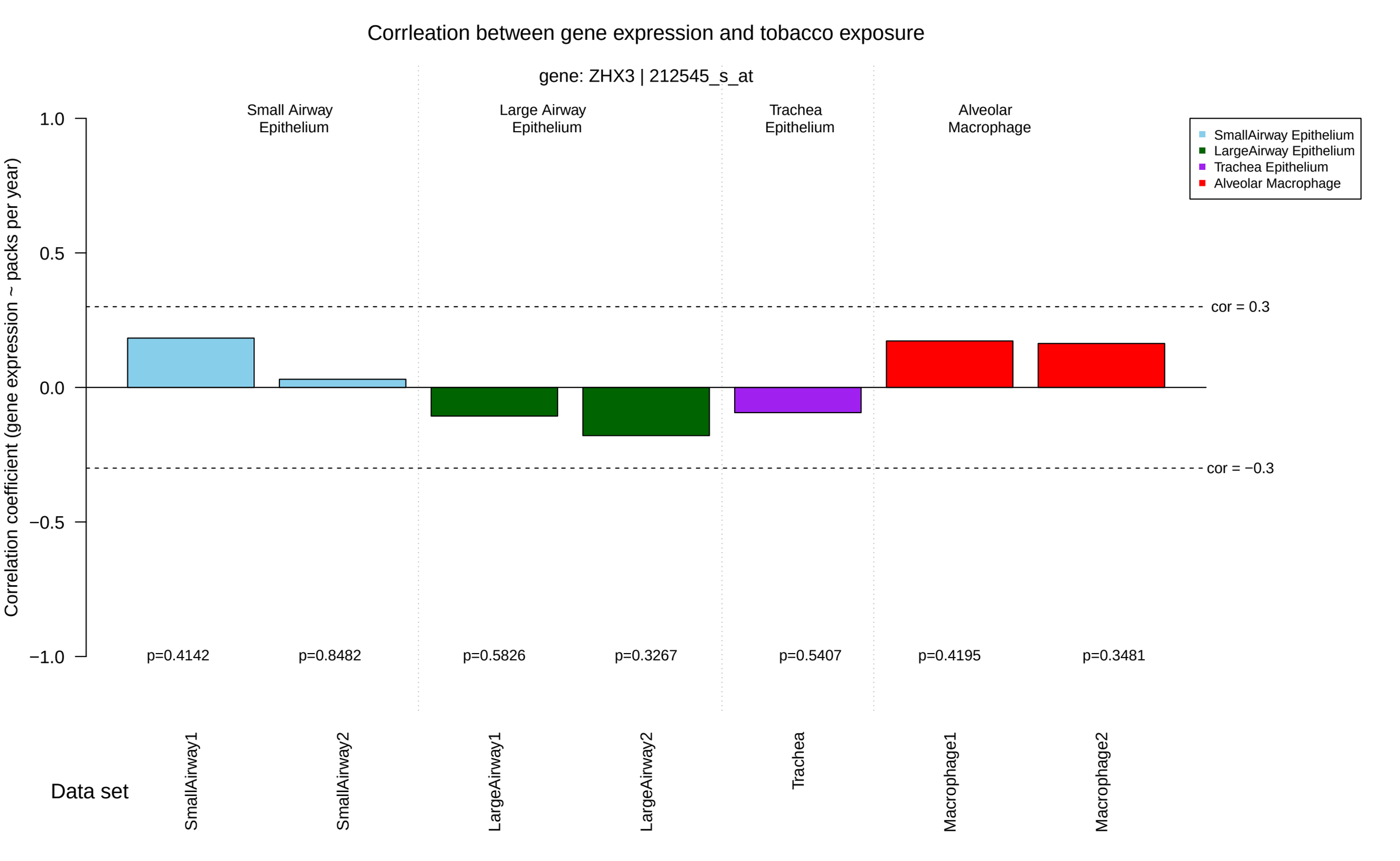

set: LargeAirway1 (p=0.4889)
ZHX3 | 212545_s_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8358)
ZHX3 | 212545_s_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.3486)
ZHX3 | 212545_s_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8106)
ZHX3 | 212545_s_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2199)
ZHX3 | 212545_s_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ZHX3 | 212545_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4142
p=0.8482
p=0.5826
p=0.3267
p=0.5407
p=0.4195
p=0.3481
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2